

Probability
Chapter 3 - Module 4

Poisson Random Variables

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Theorem. (Central Limit Theorem) Let $X_1, X_2, \dots$ be iid random variables with mean μ and variance σ^2 . Then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma} \rightarrow N(0, 1)$$

in distribution, which means that for any real number a that

$$P\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma} \leq a\right) \rightarrow \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-x^2} dx.$$

There are many extensions of this theorem including the following.

Theorem. Let $X_1, X_2, \dots$ be independent random variables satisfying

$$E[X_i] = 0, \quad \text{Var}(X_i) = \sigma_i^2, \quad E[|X_i^3|] < \infty,$$

such that

$$\frac{1}{\sigma(n)^3} \sum_{i=1}^n E[|X_i^3|] \rightarrow 0 \text{ as } n \rightarrow \infty,$$

where

$$\sigma(n)^2 = \text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \sigma_i^2.$$

Then

$$\frac{1}{\sigma(n)} \sum_{i=1}^n X_i \rightarrow N(0, 1)$$

in the sense that the CDF of $\frac{1}{\sigma(n)} \sum_{i=1}^n X_i$ converges to the CDF of the normal density function with mean zero and variance 1.

This is used to explain why so many data sets appear to be well approximated by the normal distribution. And with a few tricks we were able to approximate probabilities from the binomial distribution.

But the normal density function is not in general suitable as a model for a random variable that takes on the values $0, 1, \dots$ with probability one. But this situation is common for example when we are discussing the number of people, requests or transactions for example that arrive over a specific interval of time.

So as in Module 5 from Chapter 2 let X be a binomial random variable with parameters n and p . And define $\lambda = np$. Then when n is large and p is small

$$P(X = i) \approx e^{-\lambda} \frac{\lambda^i}{i!}$$

and so in this case the binomial random variable can be approximated by the Poisson random variable.

The Poisson distribution is a valuable tool for modeling the number of arrivals, transactions or requests that occur over some interval of time. And is a natural choice given its close relationship to the binomial distribution.

Let's look at a Mathematica Notebook to illustrate this.

Def. (Poisson Random Variable) We say that X is a Poisson random variable with parameter λ if

$$p(i) \equiv P(X = i) = e^{-\lambda} \frac{\lambda^i}{i!}$$

for $i = 0, 1, \dots$ and zero otherwise.

Recall that this is a probability mass function since

$$\sum_{i=0}^{\infty} p(i) = \sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{-\lambda} e^{\lambda} = 1.$$

We can also compute the expected value as follows:

$$\begin{aligned} E[X] &= \sum_{i=0}^{\infty} i p(i) \\ &= \sum_{i=0}^{\infty} i e^{-\lambda} \frac{\lambda^i}{i!} \\ &= e^{-\lambda} \sum_{i=1}^{\infty} i \frac{\lambda^i}{i!} \\ &= \lambda e^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ &= \lambda. \end{aligned}$$

We could also get higher order moments in this way, but let's use the moment generating function instead.

$$\begin{aligned}M(t) &= \sum_{i=0}^{\infty} e^{it} e^{-\lambda} \frac{\lambda^i}{i!} \\&= e^{-\lambda} \sum_{i=0}^{\infty} e^{it} \frac{\lambda^i}{i!} \\&= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!} \\&= e^{-\lambda} e^{(\lambda e^t)} \\&= e^{(\lambda(e^t - 1))}.\end{aligned}$$

Given that

$$M(t) = e^{(\lambda(e^t - 1))}$$

we can first see that

$$M(0) = e^{(\lambda(e^0 - 1))} = 1.$$

Since

$$M'(t) = e^{(\lambda(e^t - 1))} \lambda e^t$$

we have that the first moment is

$$M'(0) = \lambda.$$

And since

$$M''(t) = e^{(\lambda(e^t - 1))} (\lambda e^t)^2 + e^{(\lambda(e^t - 1))} \lambda e^t$$

we have that the second moment is

$$M''(0) = \lambda^2 + \lambda.$$

Finally we conclude that

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

So the mean and the variance are equal.

There are many interesting properties.

We first note that if

X_1 is a Poisson random variable with mean λ_1 and

X_2 is a Poisson random variable with mean λ_2 and if

X_1 and X_2 are independent,

then $Y = X_1 + X_2$ is Poisson random variable with mean $\lambda_1 + \lambda_2$.

This follows immediately from the fact that the moment generating function Y is

$$\begin{aligned} M(t) = M_1(t) M_2(t) &= e^{\lambda_1(e^t-1)} e^{\lambda_2(e^t-1)} \\ &= e^{(\lambda_1+\lambda_2)(e^t-1)}. \end{aligned}$$

This extends using induction to the following:

Theorem. Suppose that $X_1, X_2, \dots, X_m$ are independent Poisson random variables with means $\lambda_1, \lambda_2, \dots, \lambda_m$ respectively. Then $Y = \sum_{i=1}^m X_i$ is a

Poisson random variable with mean $\sum_{i=1}^m \lambda_i$.

Example 1. Suppose that the number of people in a class is a Poisson random variable with mean λ . And suppose the the gender of each person that comes to class is an independent random variable with the probability that it is a women is p and the probability that it is a man is $(1 - p)$.

Let N_1 and N_2 be the number of women and men respectively that come to class.

Problem. Determine

$$P(N_1 = n, N_2 = m).$$

Solution. Let $N = N_1 + N_2$.We solve the problem by conditioning on N .

So note that

$$\begin{aligned}P(N_1 = n, N_2 = m) &= \sum_{i=0}^{\infty} P(N_1 = n, N_2 = m \mid N = i) P(N = i) \\&= P(N_1 = n, N_2 = m) P(N = n + m) \\&= \left[\binom{n+m}{n} p^n (1-p)^m \right] \left[e^{-\lambda} \frac{\lambda^{n+m}}{(n+m)!} \right] \\&= \frac{(n+m)!}{n! m!} p^n (1-p)^m e^{-\lambda p} e^{-\lambda(1-p)} \frac{\lambda^n \lambda^m}{(n+m)!} \\&= \left[e^{-\lambda p} \frac{(\lambda p)^n}{n!} \right] \left[e^{-\lambda(1-p)} \frac{(\lambda(1-p))^m}{m!} \right]\end{aligned}$$

So N_1 is a Poisson random variable with mean λp and

N_2 is a Poisson random variable with mean $\lambda(1-p)$

and N_1 and N_2 are independent.

Summary.

Suppose that the number of arrivals is a Poisson random variable with mean λ .

And suppose that the arrivals are classified into two classes C_1 and C_2 with probability p and $1 - p$ respectively and that these classifications are independent.

Then N_1 , the number of arrivals classified into C_1 and N_2 , the number of arrivals that are classified into C_2 are independent Poisson random variables with mean λp and $\lambda(1 - p)$ respectively.

This extends further, but first we need to consider the multinomial distribution.

You will recall that if X is a binomial random variable with parameters p and n that

$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}.$$

To facilitate our translation, we write $n = n_1 + n_2$, where n_1 is the number of successes and n_2 is the number of failures.

In addition let's write $p_1 = p$ and $p_2 = 1 - p$, where $p_1 + p_2 = 1$.

Then

$$P(X = n_1) = \binom{n_1 + n_2}{n_1} p^{n_1} p_2^{n_2} = \frac{(n_1 + n_2)!}{n_1! n_2!} p_1^{n_1} p_2^{n_2}$$

or to improve the symmetry we could let X_1 denote the random number of successes and X_2 denote the random number of failures. Then

$$P(X_1 = n_1, X_2 = n_2) = \frac{(n_1 + n_2)!}{n_1! n_2!} p_1^{n_1} p_2^{n_2}.$$

Def. (Multinomial Distribution) Consider n experiments with outcomes that are classified into m distinct classes C_i with probability $p_i \geq 0$.

And let X_i denote the random number of outcomes that are classified into class C_i for $i = 1, \dots, m$. And suppose that the classifications are independent events.

Then

$$P(X_1 = n_1, X_2 = n_2, \dots, X_m = n_m) = \frac{(n_1 + n_2 + \dots + n_m)!}{n_1! n_2! \dots n_m!} p_1^{n_1} p_2^{n_2} \dots p_m^{n_m}$$

and we say that $X_1, X_2, \dots, X_n$ have a multinomial distribution for $n = n_1 + \dots + n_m$.

Note that in the case that $p_i = 0$ for any i you should interpret $p_i^{n_i}$ as 1 whenever $n_i = 0$.

Theorem. Suppose that the number of arrivals is a Poisson random variable with mean λ .

Suppose that each arrival is classified into m classes $C_1, C_2, \dots, C_m$ with probability $p_1, p_2, \dots, p_m$ respectively, where $p_i \geq 0$ for each i and $\sum_{i=1}^m p_i = 1$.

Suppose that the classifications are all independent.

Let N_i denote the random number of arrivals that are put in class C_i .

Then $N_1, N_2, \dots, N_m$ are independent Poisson random variables with

$$P(N_i = n_i) = e^{-\lambda p_i} \frac{(\lambda p_i)^{n_i}}{n_i!}.$$

So then

$$\begin{aligned} P(N_1 = n_1, N_2 = n_2, \dots, N_m = n_m) &= \prod_{i=1}^m P(N_i = n_i) \\ &= \prod_{i=1}^m e^{-\lambda p_i} \frac{(\lambda p_i)^{n_i}}{n_i!}. \end{aligned}$$

Proof. As before we condition on $N = N_1 + \dots + N_m$. Then

$$\begin{aligned}
 & P(N_1 = n_1, N_2 = n_2, \dots, N_m = n_m) \\
 &= \sum_{i=0}^{\infty} P(N_1 = n_1, N_2 = n_2, \dots, N_m = n_m \mid N = i) P(N = i) \\
 &= P\left(N_1 = n_1, N_2 = n_2, \dots, N_m = n_m \mid N = \sum_{j=1}^m n_j\right) P\left(N = \sum_{j=1}^m n_j\right) \\
 &= P(N_1 = n_1, N_2 = n_2, \dots, N_m = n_m) P\left(N = \sum_{j=1}^m n_j\right) \\
 &= \left[\frac{(n_1 + \dots + n_m)!}{n_1! n_2! \dots n_m!} p_1^{n_1} p_2^{n_2} \dots p_m^{n_m} \right] \left[e^{-\lambda} \frac{\lambda^{n_1 + \dots + n_m}}{(n_1 + \dots + n_m)!} \right] \\
 &= e^{(-\lambda p_1)} \frac{(\lambda p_1)^{n_1}}{n_1!} \dots e^{(-\lambda p_m)} \frac{(\lambda p_m)^{n_m}}{n_m!},
 \end{aligned}$$

where we have used the fact that $\lambda = \lambda \sum_{i=1}^m p_i = \sum_{i=1}^m \lambda p_i$.